Video-MoFlow: Video-Conditioned One-Step Flow Matching for Human Trajectory Forecasting

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Abstract

Human trajectory forecasting requires modeling complex, multi-modal future movements. We extend the one-step flow matching model, MoFlow, by integrating rich visual evidence from video inputs. By conditioning the generative flow field on global visual tokens, our proposed Video-MoFlow significantly reduces spatial ambiguity. Preliminary results on the ETH/UCY dataset demonstrate the efficacy of our global video encoder in improving trajectory precision.

1 Introduction

Current generative models for trajectory prediction, such as the Conditional Flow Matching model (MoFlow) [**?**], rely strictly on historical kinematic data to predict future coordinates. However, past trajectories are often insufficient to explain future movements when environmental structures or obstacles constrain an agent's path.

In this work, we propose Video-MoFlow. We formulate trajectory forecasting as a video-conditioned flow matching problem, where the future motion distribution is generated by a flow field modulated by both past kinematics and visual evidence. We specifically introduce a global video encoder to process environmental context to investigate its influence on the predicted probability paths. This report outlines the current architecture and provides an empirical evaluation of its preliminary performance.

2 Methodology

2.1 Conditional Flow Matching

The baseline MoFlow architecture approximates a target vector field U^t by learning a neural network v_θ that maps a Gaussian noise distribution to the true data distribution. For numerical stability, the model solves an equivalent data-space prediction task:

$$D_\theta(Y^t, C, t) := Y^t + (1-t)v_\theta(Y^t, C, t) \tag{1}$$

where $t \in [0, 1)$ represents the flow time, and C represents the contextual information.

2.2 Video-Conditioned Multimodal Fusion

To extend this to a video-conditioned framework, we process an observed video clip $V \in \mathbb{R}^{T_{obs} \times H \times W \times 3}$. A global video encoder extracts scene-level context $z_{video_global} \in \mathbb{R}^{L_{global} \times D}$.

To combine this with the trajectory-social tokens z_{traj}, we employ a Cross-Attention fusion mechanism:

$$z_{fused} = \text{CrossAttention}(Q = z_{traj}, K = z_{video}, V = z_{video}) \tag{2}$$

The resulting fused context modulates the generative flow field $v_\theta(Y^t, t | z_{traj}, z_{video})$, directly influencing the sampling of the K diverse future trajectories.

2.3 Agent-Centric Visual Conditioning

In parallel to the global scene context, we also formulate an alternative agent-centric visual encoding strategy. Rather than extracting a single representation for the entire environment, this approach dynamically extracts localized visual patches from the video frames.

For each agent i at a given timestep, spatial coordinates are projected onto the pixel space using the scene's homography matrix. A visual patch $P_i \in \mathbb{R}^{H_p \times W_p \times 3}$ is cropped around the agent's mapped location. A dedicated convolutional encoder then processes these patches to yield agent-specific visual tokens $z^{(i)}_{video_local} \in \mathbb{R}^D$.

This localized formulation aims to capture fine-grained environmental constraints immediately surrounding the pedestrian (e.g., adjacent obstacles or physical boundaries). The resulting local tokens can be fused with the agent's corresponding trajectory tokens, offering a highly localized modulation of the generative flow field. The success of this approach is strictly dependent on the precision of the world-to-pixel coordinate transformations.

2.4 Architecture Overview

The overall processing pipeline of Video-MoFlow is illustrated in Figure 1.

3 Experiments and Results

To evaluate the impact of visual conditioning, we compare the global video encoder implementation of Video-MoFlow against the baseline MoFlow model on the ETH/UCY dataset.

3.1 Quantitative Results

Our evaluation focuses on the minimum Average Displacement Error (minADE_{20}) at 4.8s and the minimum Final Displacement Error (minFDE_{20}) at 4.0s. As shown in Table 1, the current integration of global scene context generally increases displacement errors compared to the baseline.Performance degradation is most notable on the hotel and eth subsets, with minADE errors increasing by 176.95% and 20.18%, respectively. Conversely, marginal performance gains are observed on the zara1 subset (+6.97% in minADE; +13.95% in minFDE). These preliminary findings indicate that while the architecture executes successfully, the global visual features may currently introduce noise

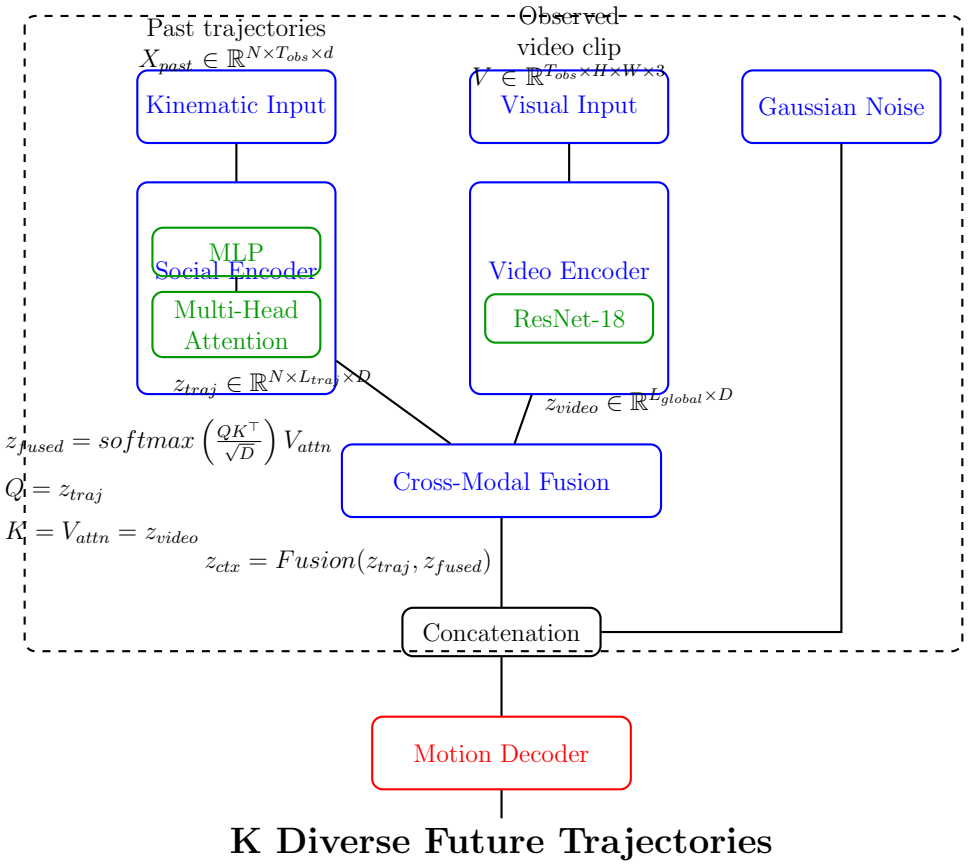


Figure 1: Overview of the Video-MoFlow architecture. Global visual tokens and social-trajectory tokens are fused to modulate the conditional flow matching decoder.

or unaligned contextual priors. This necessitates further investigation and debugging of the visual-kinematic cross-attention layers.

Table 1: Quantitative comparison on the ETH/UCY dataset. Lower values indicate better performance.

Dataset	Metric	Horizon	MoFlow (Teacher)	GVM (Teacher)	Performance Gain
eth	ADE_min	4.8s	0.462023	0.555257	-20.18%
eth	FDE_min	4.0s	0.666986	0.826500	-23.92%
hotel	ADE_min	4.8s	0.152223	0.421575	-176.95%
hotel	FDE_min	4.0s	0.268332	0.765176	-185.16%
univ	ADE_min	4.8s	0.247981	0.292342	-17.89%
univ	FDE_min	4.0s	0.438123	0.540382	-23.34%
zara1	ADE_min	4.8s	0.172554	0.160521	+6.97%
zara1	FDE_min	4.0s	0.309535	0.266357	+13.95%
zara2	ADE_min	4.8s	0.124809	0.126629	-1.46%
zara2	FDE_min	4.0s	0.223136	0.214746	+3.76%

4 Implementation Roadblocks and Diagnostic Status

The integration of visual encoders has introduced specific data pipeline and coordinate mapping challenges. Current development is focused on isolating these bottlenecks across both the global and agent-centric implementations.

4.1 Global Video Encoder: Stanford Drone Dataset (SDD)

Hardware and Memory Constraints: Development on the SDD dataset is currently paused due to compute limitations. The ingestion pipeline for the custom 70 GB SDD repository (comprising raw video scenes and raw text annotations) triggered an Out-Of-Memory (OOM) failure, resulting in a crash on the remote compute node. Active debugging and memory optimization will resume once access to the remote environment is restored.

Evaluation Protocol Discrepancies: Adhering strictly to the Leave-One-Scene-Out (LOSO) cross-validation protocol requires significant refactoring of the baseline evaluation logic, as the default MoFlow repository does not natively execute this protocol for SDD. Furthermore, metric evaluation on SDD is conducted in pixel space, bypassing the homography requirements of ETH/UCY, which necessitates a parallel, isolated evaluation pipeline.

4.2 Agent-Centric Video Encoder

Coordinate Mapping Failures: The agent-centric visual pipeline is currently failing due to inaccurate world-to-pixel coordinate mapping. Visual diagnostics confirm that the dynamic cropping mechanism frequently extracts spatial patches that do not contain the target agents.

Homography and Projection Bugs: This failure is classified as a severe logic bug stemming from misaligned homography matrices. Preliminary testing indicates that

while the transformation matrices (sourced via OpenTraj) project trajectories accurately for the *zara1* and *zara2* scenes, they produce significant spatial drift on other subsets such as *eth* and *hotel*. Consequently, the visual tokens extracted do not correlate spatially with the kinematic data, corrupting the feature fusion process. Precise isolation of the matrix transformations is pending the restoration of compute access.

5 Next Steps and Action Plan

To address the current implementation roadblocks and advance the evaluation of Video-MoFlow, the immediate action plan is structured around isolated debugging and external pipeline development. Activities are temporarily scoped to accommodate the current lack of access to the remote computing environment.

5.1 Homography Validation and Correction

To resolve the coordinate mapping failures in the agent-centric encoder, the immediate priority is to rigorously validate the OpenTraj homography matrices for the ETH and Hotel subsets. We will construct an isolated, standalone local debugging script using OpenCV to manually project trajectory coordinates onto sample frames. This will allow us to quantify the spatial drift and formulate corrected transformation matrices before re-integrating them into the pipeline.

5.2 SDD Evaluation Pipeline Refactoring

To adhere to the required Leave-One-Scene-Out (LOSO) cross-validation protocol for the Stanford Drone Dataset, we will develop dedicated, external evaluation scripts. This approach ensures we can interact with the core modeling repository to compute pixel-space metrics without modifying any lines of its underlying source code.

5.3 Memory Optimization

Upon restoration of remote compute access, we will implement memory-efficient data loading strategies for the 70 GB SDD repository. This will involve investigating dynamic batch loading and tensor offloading techniques to prevent the Out-Of-Memory failures previously encountered during the scene ingestion phase.